

Continuous Assessment Test (CAT) – I - August 2026

Programme	M.Tech.	Semester	Fall Semester – 2026-27
Course Code & Course Title	MACSE512 & Operating Systems	Class Number	CH2026270103464
Faculty	Dr.Kumar R	Slot	DI+TDI
Duration	90 Minutes	Max. Mark	50

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information
- Only non-programmable calculator without storage is permitted

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	BT Level
1	a.	An automotive technology firm is developing a real-time central control unit for an Autonomous Electric Vehicle (EV). The system must rapidly process multi-sensor inputs (LiDAR, radar, brake-by-wire sensors), execute low-latency motor control algorithms and manage vehicle dynamics. To guarantee passenger safety, the platform must be highly fault-tolerant: critical safety services (emergency braking, active steering, collision avoidance) must remain operational even if non-critical applications (infotainment, cabin climate controls, trip logging, diagnostic telemetry) crash or fail. The engineering team is evaluating which Operating System (OS) structuring approach to adopt.	10	CO1	K2
	b.	Discuss the merits and demerits of implementing various OS structuring methods (e.g., Monolithic, Layered, Modular) in the context of the given autonomous vehicle scenario. (5 marks)			
		During a test drive, a memory leak causes the cabin climate-control and infotainment module to crash. Recommend an OS structure for the central control unit and justify your choice by detailing the exact hardware and software mechanisms that ensure emergency braking and active steering remain completely unaffected by this crash. (5 marks)			
2		<pre>#include <sys/types.h> #include <stdio.h> #include <unistd.h> int count = 500; int main()</pre>			

	<pre> { pid_t pid; pid = fork(); if (pid == 0) count += 150; return 0; } else if (pid > 0) wait(NULL); printf("PARENT: count = %d",count); return 0; } } </pre> a. What is the final value of the variable 'count' of the parent process? Justify your answer in detail. (2 Marks) b. Devise a multithreaded program in 'C' for the above code snippet using POSIX APIs. (6 Marks) c. What happens if you use threads (pthread_create()) instead of fork () system calls and determine the final value of the 'count' variable. (2 Marks)	10	CO1	K3
3	<pre> #include <stdio.h> #include <unistd.h> int main() { int i; for (i =0;i < 3;i++) fork(); return 0; } </pre> a. How many processes are created? Draw a process tree diagram showing how all the child processes branch out. (4 Marks) b. Explain how to use wait() system call to properly clean up these all the child processes and prevent zombie states (2 Marks) c. How does an orphan process differ from a zombie process? Illustrate your answer with examples. (4 Marks)	10	CO2	K3
4	Consider a system running ten I/O-bound tasks and one CPU-bound task. Assume that the I/O-bound tasks issue an I/O operation once for every millisecond of CPU computing and that each I/O operation takes 10 milliseconds to			

		complete. Also assume that the context-switching overhead is 0.1 millisecond and that all processes are long-running tasks. Determine the CPU utilization for a round-robin scheduler when: a. The time quantum is 1 millisecond (5 Marks) b. The time quantum is 10 milliseconds (5 Marks)	10	CO2	K3																								
5		VIT health center's emergency care unit must efficiently allocate doctors' time to treat incoming patients based on their arrival time, required treatment duration, and urgency. The health center follows two different scheduling strategies: a. The patient requiring the least remaining treatment time is always given preference. (5 marks) b. Each patient receives a fixed treatment slot of 3 time units. If multiple patients arrive at the same time, the one with the higher priority is treated first in the initial cycle. However, after receiving their first 3 time units, they are placed at the back of the queue, and further treatment follows the order of the queue, regardless of priority. (5 marks) Draw the treatment timeline and calculate the Average Waiting Time and Average Turnaround Time for each strategy. Finally, analyze the results to decide which strategy works better. The patient details are given below. <table border="1"><thead><tr><th>Patient ID</th><th>Arrival Time</th><th>Required Treatment time</th><th>Priority</th></tr></thead><tbody><tr><td>P1</td><td>0</td><td>10</td><td>2</td></tr><tr><td>P2</td><td>2</td><td>4</td><td>4</td></tr><tr><td>P3</td><td>3</td><td>8</td><td>1</td></tr><tr><td>P4</td><td>5</td><td>6</td><td>3</td></tr><tr><td>P5</td><td>7</td><td>5</td><td>2</td></tr></tbody></table>	Patient ID	Arrival Time	Required Treatment time	Priority	P1	0	10	2	P2	2	4	4	P3	3	8	1	P4	5	6	3	P5	7	5	2	10	CO2	K2
Patient ID	Arrival Time	Required Treatment time	Priority																										
P1	0	10	2																										
P2	2	4	4																										
P3	3	8	1																										
P4	5	6	3																										
P5	7	5	2																										
***** All the best *****																													